

BE 305: Bioethics and Regulatory Affairs

Course Code	BE 305
Course Name	Bioethics and Regulatory Affairs
L-T-P-C	1-0-0-1
Prerequisites	IC136
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	44th BoA (Nov 2021); 37th Senate (Oct 2022), in the B.Tech Bioengineering curriculum

Course Contents

- **Introduction to Bioethics**

Ethics and ethical theory, the nuremberg code, declaration of helsinki, the belmont report, history of ethics, justice and rights, liberty and morality. Deontology, utilitarianism and Principalism, Virtue ethics, Ethics of Care, Human Rights. [4 Lectures]

- **Ethics of research involving human participants**

Biomedical research involving human participants. Guidelines for research on children, Stem cell research, animal research etc. Ethical issues in Genetic research and studies involving Genetic information. Ethics of investigator-participant relationship, problems of randomized clinical trial, constitution of Ethics Committees, informed consent and its theoretical value (respecting autonomy, nonmaleficence, concept of confidentiality etc.), research on vulnerable population. [5 Lectures]

- **Drug regulatory affairs**

Pharmaceutical regulatory affairs, national regulatory affairs, drug approval and international drug regulations, regulatory affairs of controlled drug delivery system, regulatory requirements for product approvals, environmental concerns and regulations. National regulatory affairs and bodies – Central Drugs Standard Control Organization (CDSCO) headed by the Drug Controller General of India (DCGI). [5 Lectures]

Text Books

1. Ronald Munson, **Intervention and Reflection: Basic Issues in Medical Ethics**, 10th Edition, Cengage Learning, 2018.
2. Beauchamp, Tom L and Childress, James F., **Principles of Biomedical Ethics**, Oxford University Press, 2019.

References

1. Rachels, James and Stuart Rachels, **Elements of Moral Philosophy**, 5-7th editions, McGraw-Hill, 2012.
2. Richard Durbin, Sean R. Eddy, Anders Krogh, Graeme Mitchison, **Biological Sequence Analysis- Probabilistic Models of Proteins and Nucleic Acids**, Cambridge University Press.
3. NSantoro, Michael A and Thomas M Gorrie, **Ethics and the Pharmaceutical Industry**, Cambridge University Press.